Permutations & Combinations

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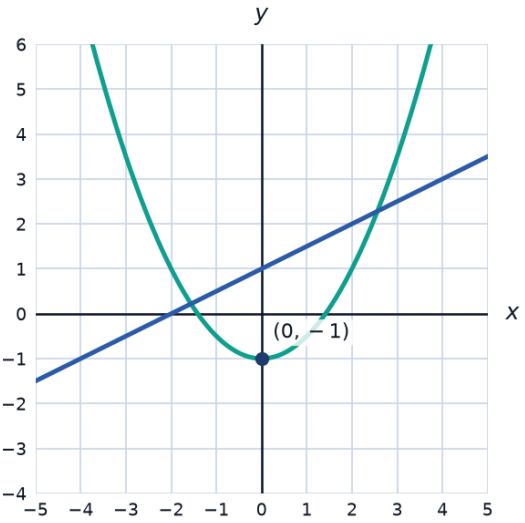
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Fundamental Counting & Factorials**

A committee of 3 people is to be selected from a group of 5 people. The number of ways to form this committee is given by the combination formula ${}^{n}C_{r} = \frac{n!}{r!\,(n-r)!}$. How many different committees can be formed? Show your complete algebraic work.

2 **[Foundational]** • **Target: Permutations with Repetition & Restrictions**

Find the number of distinct arrangements of the letters of the word COMMITTEE.



3 **[Foundational]** • **Target: Combinations & Committee Selection**

Find the number of distinct arrangements of the letters of the word SUCCESSFULLY.

4 **[Foundational]** • **Target: Complex Multi-Condition Problems**

Find the number of distinct arrangements of the letters of the word ENGINEERING.

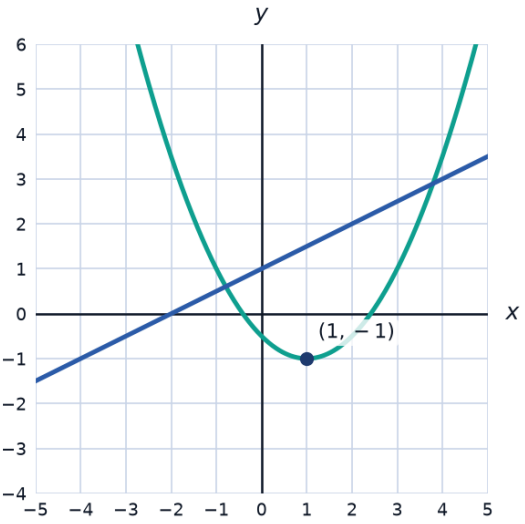
5 **[Foundational]** • **Target: Fundamental Counting & Factorials**

Find the number of distinct arrangements of the letters of the word ARRANGEMENT.

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6 [Foundational] • Target: Permutations with Repetition & Restrictions

Find the number of distinct arrangements of the letters of the word MATHEMATICS.



7 [Foundational] • Target: Combinations & Committee Selection

Find the number of distinct arrangements of the letters of the word ACCOUNTANT.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Complex Multi-Condition Problems

Evaluate ${}^{7}P_{2}$.

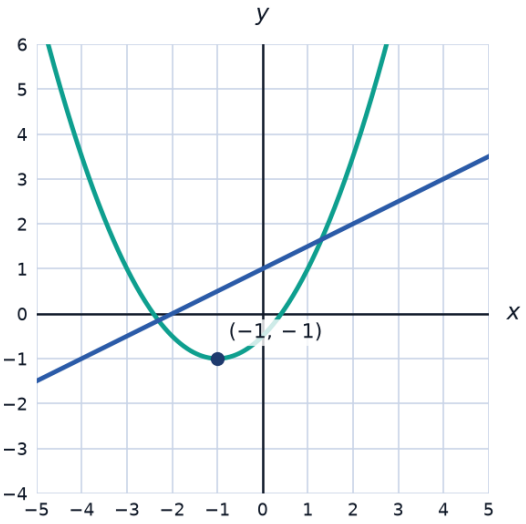
9 [Intermediate] • Target: Fundamental Counting & Factorials

Evaluate ${}^{5}P_{3}$.

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10 [Intermediate] • Target: Permutations with Repetition & Restrictions

Evaluate ${}^{6}P_{2}$.



11 [Intermediate] • Target: Combinations & Committee Selection

Evaluate ${}^{9}P_{2}$.

12 [Intermediate] • Target: Complex Multi-Condition Problems

Evaluate ${}^{4}P_{2}$.

13 [Intermediate] • Target: Fundamental Counting & Factorials

Evaluate ${}^{5}P_{2}$.

14 [Intermediate] • Target: Permutations with Repetition & Restrictions

In how many distinct ways can 12 different people be seated around a circular table (rotations considered identical)? Show your complete algebraic work.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

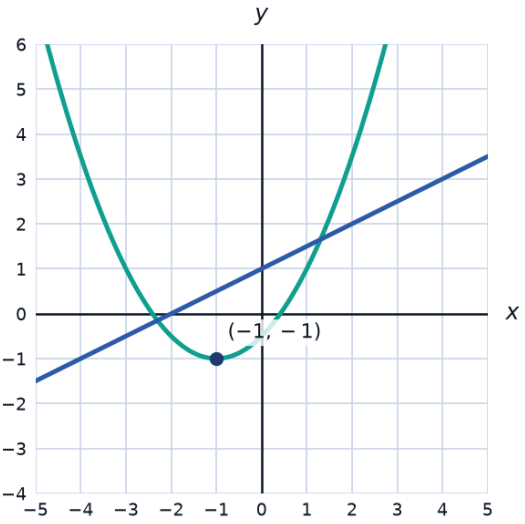
15 [Challenging] • Target: Combinations & Committee Selection • Cross-Lesson: 12.01 Forces & Equilibrium

Find the number of distinct arrangements of the letters of the word MISSISSIPPI.

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16 [Challenging] • Target: Complex Multi-Condition Problems • Cross-Lesson: 13.01 Newton's Laws

Find the number of distinct arrangements of the letters of the word STATISTICS.



17 [Challenging] • Target: Fundamental Counting & Factorials • Cross-Lesson: 14.01 Energy, Work & Power

In how many distinct ways can 11 different people be seated around a circular table (rotations considered identical)? Show your complete algebraic work.

18 [Challenging] • Target: Permutations with Repetition & Restrictions • Cross-Lesson: 15.01 Representation of Data

In how many distinct ways can 10 different people be seated around a circular table (rotations considered identical)? Show your complete algebraic work.

19 [Challenging] • Target: Combinations & Committee Selection • Cross-Lesson: 17.01 Discrete Random Variables

Evaluate $3!$.

20 [Challenging] • Target: Complex Multi-Condition Problems • Cross-Lesson: 18.01 Binomial Distribution

Find the number of distinct arrangements of the letters of the word ARRANGE.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Fundamental Counting & Factorials

How many 4-digit numbers can be formed using the digits $1, 2, 3, 4, 5, 6$ without repetition?

a. 360

b. 720

c. 120

d. 240

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22 [Foundational] • Target: Fundamental Counting & Factorials

In how many ways can 7 runners finish 1st, 2nd, and 3rd in a race, assuming no ties?

a. 210

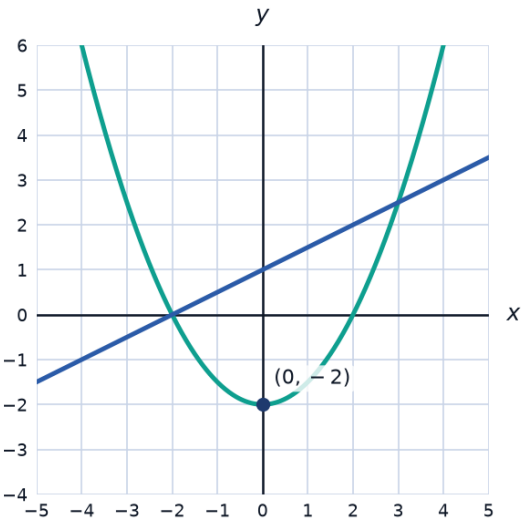
b. 35

c. 5040

d. 120

23 [Foundational] • Target: Fundamental Counting & Factorials

How many even 3-digit numbers can be formed using digits $1, 2, 3, 4, 5$ without repetition?



a. 24

b. 12

c. 36

d. 60

24 [Foundational] • Target: Permutations with Repetition & Restrictions

How many distinct arrangements of all the letters in the word MATHEMATICS are possible?

a. $4,989,600$

b. $39,916,800$

c. $1,247,400$

d. $2,494,800$

25 [Foundational] • Target: Permutations with Repetition & Restrictions

In how many ways can 5 boys and 4 girls stand in a line if all 4 girls must stand together?

a. $17,280$

b. $2,880$

c. $43,200$

d. $5,760$

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26 [Foundational] • Target: Permutations with Repetition & Restrictions

In how many ways can 6 people sit around a circular table, where rotations are considered equivalent?

a. 120

b. 720

c. 24

d. 360

27 [Foundational] • Target: Combinations & Committee Selection

A committee of 5 is to be chosen from a group of 8 men and 6 women. How many committees are possible if it must contain exactly 3 men and 2 women?

a. 840

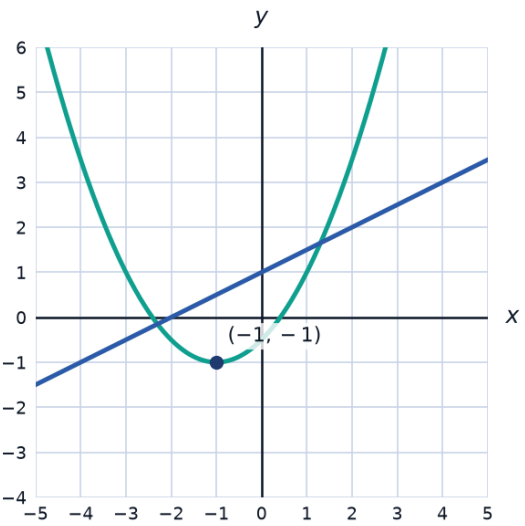
b. 420

c. 1260

d. 2002

28 [Foundational] • Target: Combinations & Committee Selection

From a group of 10 students, a team of 4 is selected. If the team must include the student council president, how many distinct teams are possible?



a. 84

b. 210

c. 126

d. 72

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Combinations & Committee Selection

A team of 4 is selected from 7 boys and 5 girls. How many teams have at least one girl?

a. 460

b. 495

c. 420

d. 35

30 [Intermediate] • Target: Complex Multi-Condition Problems

How many 5-digit odd numbers greater than 40,000 can be formed using digits $2, 3, 4, 5, 7$ without repetition?

a. 28

b. 36

c. 48

d. 24

31 [Intermediate] • Target: Complex Multi-Condition Problems

A delegation of 4 is chosen from 5 teachers and 6 students. In how many ways can this be done if there must be more teachers than students?

a. 75

b. 65

c. 85

d. 50

32 [Intermediate] • Target: Complex Multi-Condition Problems

Four cards are drawn from a standard 52-card deck. What is the probability that all four cards are of the same suit?

a. $\frac{44}{4165} \approx 0.01056$

b. 0.0025

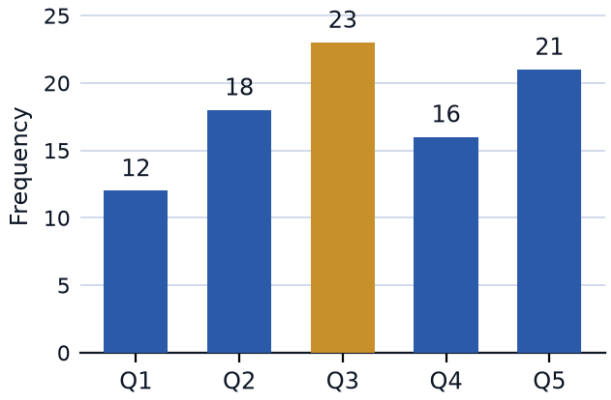
c. 0.0450

d. 0.0210

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33 [Intermediate] • Target: Fundamental Counting & Factorials

In how many distinct ways can 9 different people be seated around a circular table (rotations considered identical)?



a. 38504

b. 39347

c. 35536

d. 40320

34 [Intermediate] • Target: Permutations with Repetition & Restrictions

Find the number of distinct arrangements of the letters of the word BALLOON.

a. 1076

b. 1077

c. 1244

d. 1260

35 [Intermediate] • Target: Combinations & Committee Selection

Find the number of distinct arrangements of the letters of the word ADDRESS.

a. 1146

b. 1260

c. 1205

d. 1211

36 [Intermediate] • Target: Complex Multi-Condition Problems

Evaluate $12!$.

a. 39916800

b. 479001600

c. 479001588

d. 6227020800

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37 [Intermediate] • Target: Fundamental Counting & Factorials

In how many distinct ways can 8 different people be seated around a circular table (rotations considered identical)?

a. 5561

b. 5769

c. 5040

d. 5466

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Permutations with Repetition & Restrictions • Cross-Lesson: 15.01 Representation of Data

In how many distinct ways can 6 different people be seated around a circular table (rotations considered identical)?

a. 120

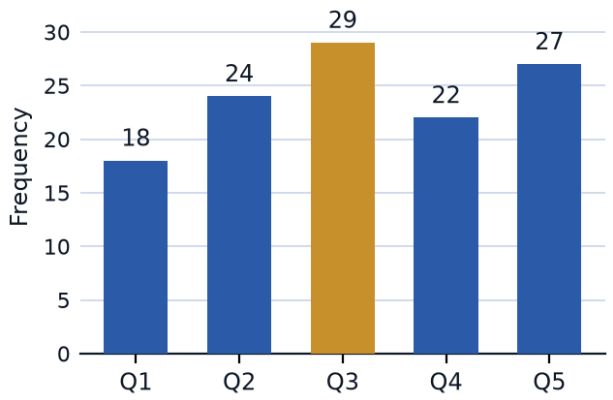
b. 103

c. 122

d. 100

39 [Challenging] • Target: Combinations & Committee Selection • Cross-Lesson: 17.01 Discrete Random Variables

In how many distinct ways can 7 different people be seated around a circular table (rotations considered identical)?



a. 720

b. 766

c. 647

d. 611

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40 [Challenging] • Target: Complex Multi-Condition Problems • Cross-Lesson: 18.01 Binomial Distribution

Find the number of distinct arrangements of the letters of the word GIGGLE.

a. 214

b. 141

c. 180

d. 207

41 [Challenging] • Target: Fundamental Counting & Factorials • Cross-Lesson: 19.01 Normal Distribution

Find the number of distinct arrangements of the letters of the word LEVEL.

a. 12

b. 29

c. 30

d. 76

42 [Challenging] • Target: Permutations with Repetition & Restrictions • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

Find the number of distinct arrangements of the letters of the word PEPPER.

a. 33

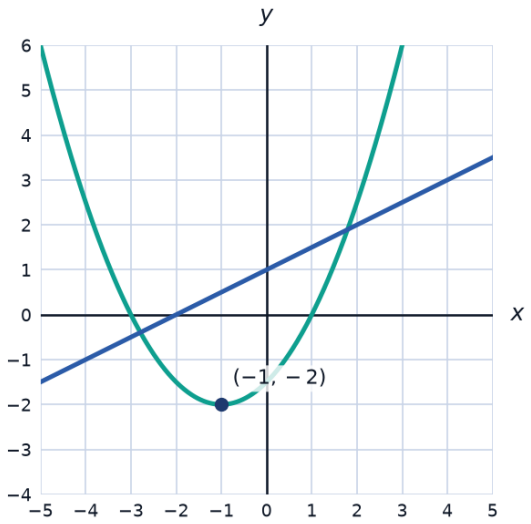
b. 17

c. 34

d. 60

43 [Challenging] • Target: Combinations & Committee Selection • Cross-Lesson: 3.01 Graphs & Transformations

Find the number of distinct arrangements of the letters of the word BANANA.



a. 52

b. 71

c. 60

d. 62

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44 [Challenging] • Target: Complex Multi-Condition Problems • Cross-Lesson: 4.01 Coordinate Geometry

Find the number of distinct arrangements of the letters of the word PUPPY.

a. 103

b. 27

c. 64

d. 60

45 [Challenging] • Target: Fundamental Counting & Factorials • Cross-Lesson: 5.01 Trigonometry

Find the number of distinct arrangements of the letters of the word ERROR.

a. 23

b. 49

c. 10

d. 39

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